TruthCheck v2

Progress report: multilingual healthcare misinformation verification

Branch `v2-integration` · report date 16 October 2026, updated 25 October 2026 · repository `F0ur-Arms/TruthCheck`

1. Summary

Update, 25 October: the main evaluation set is not a valid benchmark

A logistic regression that sees only 10 style features of the claim text (word count, punctuation, digits, hedge words) and no evidence at all scores **98.7%** on `data/final_health_claims.csv`; a bag-of-words model scores 97.9% (majority class 67.6%). Fake claims there are short phrases (4.4 words on average, 27% end with a full stop); real claims are article sentences (18 words, 98% end with a full stop). The pipeline scores in sections 1 and 5 are therefore confounded with sentence style and should be read only as a record of what was run. The public-dataset rows from SciFact, HealthVer, PUBHEALTH and ai4bharat pass the same check (style-only 0.65 vs majority 0.64; COVID-Fact fails at 0.92 vs 0.78 and is excluded). They become the interim benchmark (1,898 claims) until the reviewed golden set exists. The NLI stage results in section 6 are unaffected: they use gold evidence from the public datasets. Script: `evaluation/artifact_check.py`.

TruthCheck takes a health claim (English, Hinglish or Hindi) and returns a verdict, a confidence, a risk assessment and cited evidence. The goal of this phase was to find out how much of the v2 redesign was real, make the evaluation honest, and wire the real components into the new LangGraph pipeline.

71.5%	83.5%	0.597	65.4%
claims given a verdict (legacy: 61.5%)	accuracy on answered claims (legacy: 75.2%)	strict accuracy, abstain counted wrong (legacy: 0.462)	LLM NLI judge vs 40.4% for the old NLI model

- **Honest evaluation.** The old 0.6831 baseline hid abstentions behind a risk-score fallback and was close to what always answering "Real" would score (0.676). Abstentions are now counted and reported separately.
- **The LangGraph pipeline now runs on real components** (knowledge-base retrieval, reranking, NLI, two-axis risk, human-review pause, persistent storage). A hardcoded NLI stub is gone.
- **The weakest stage was NLI.** Three dedicated NLI models score 40 to 45% when given the correct evidence; the Qwen2.5-7B judge scores 65.4%. Swapping it in moved the full pipeline from 45.9% coverage to 71.5%.
- **The golden set is not ready.** The 3,498 rows added so far are an English candidate pool with unreviewed labels. Final model choices have to be confirmed on a reviewed golden set, especially for Hinglish and Hindi.

Read these numbers with care

All scores come from `data/final_health_claims.csv` (2,799 article-style sentences, 1,892 Real and 907 Fake), which overlaps the knowledge base. They show relative progress between pipeline versions. They are not an estimate of real-world accuracy, and they say nothing about Hindi or Hinglish. See section 8.

2. The system

```
raw input
  -> normalisation / multilingual preprocessing        (src/preprocessor.py, multilingual/)
  -> claim routing + decomposition (LLM, heuristic fallback)   (src/claims_processor.py)
       medical-advice request  =>  NOT_A_FACT_CHECK
  -> per sub-claim: multi-lane queries, BM25 + dense retrieval, RRF, cross-encoder rerank
  -> per passage: NLI judgment + source tier + quality score   (src/evidence_engine.py)
  -> calibrated confidence and verdict                         (src/calibration.py)
  -> combine sub-claim verdicts                                (graph/nodes.py)
  -> two-axis risk (confidence vs. harm)                       (src/risk_engine_v2.py)
  -> high harm or medication-stopping claim: pause for a human (LangGraph interrupt)
  -> response with citations
```

Models. BAAI/bge-m3 (embeddings), BAAI/bge-reranker-v2-m3 (reranker), Qwen2.5-7B-Instruct-AWQ served by vLLM on the IIITD CB GPU cluster (claim decomposition and, now, the NLI judge). The old mDeBERTa NLI model remains as a fallback. **Verdicts:** SUPPORTED, SUPPORTED_WITH_CAVEATS, MIXED_EVIDENCE, MISLEADING, MOSTLY_FALSE, FALSE, INSUFFICIENT_EVIDENCE, NOT_A_FACT_CHECK.

Two entry points exist: the legacy `main.py` and the LangGraph app in `graph/` served by `app/api/main.py`. They now share one evidence engine.

3. Where we started

The audit of the repository against the v2 redesign document found that the redesign document described more than the code did.

- The LangGraph evidence node used a hardcoded NLI stub (`"SUPPORTS" if "warm water" in text`).
- The human-review node approved everything automatically; confidence weights were hand-set; the harm score was not used.
- The golden-set candidates were English-only and overlapped the evaluation set.
- The reported baseline (accuracy 0.6831, F1 0.6709) was produced without the LLM connected and decided uncertain claims from a risk-score fallback.
- The repository did not install cleanly (unresolvable spaCy model pin, Python 3.13 vs 3.11).

4. What was done

Step 0: Housekeeping

- Made the repo installable on Python 3.11 (fixed the spaCy model pin, rewrote `.gitignore`, removed stray files).
- Rewrote the README and removed config constants that nothing used.
- Test suite brought to a passing state (now 78 tests).

Step 1: Honest evaluation

- `get_binary_prediction` now returns Real, Fake, or *abstain* instead of silently falling back to a risk threshold.
- New `src/binary_scoring.py` reports coverage, strict accuracy (abstain counts as wrong) and accuracy on answered claims.
- `generate_evaluation_report.py` no longer invents numbers: it refuses to run without measured metrics.
- `run_validation_audit.py` renamed `smoke_audit.py` because it is a smoke test, not an evaluation.

Step 2: Golden set (in progress, with the teammate)

- Template and annotation guidelines written (`evaluation/golden_set/GUIDELINES.md`): 8 verdicts, language mix, population and cessation cases, double annotation with Cohen's kappa.
- A validator (`validate_golden.py`) now checks the file against the guidelines. On the current file it reports: verdicts outside the taxonomy (`TRUE/UNCERTAIN/PARTLY_TRUE`), 100% English, no row labelled or annotated, no specific claim types, 33 duplicates, 1 overlap with the evaluation set.
- About 2,260 rows have a claim, an evidence quote and a label, which is enough for the stage-level NLI test in step 4.

Step 3: Wire real components into the graph

- **Shared evidence engine** (`src/evidence_engine.py`): retrieval, rerank, NLI, source tier and quality scoring in one class used by both `main.py` and the graph.
- **Graph nodes** now retrieve from the 36,316-passage knowledge base for every sub-claim, run real NLI, and use source tiers read from the knowledge-base filename.
- **Risk:** the two-axis risk engine is used. Only high harm or a medication-cessation override blocks a response for a human; low confidence is answered as INSUFFICIENT_EVIDENCE and flagged.
- **Human review:** a real LangGraph `interrupt()`; the API returns `pending_review` and `POST /api/v1/review/{id}` resumes the same thread.
- **Persistence:** SQLite checkpointer plus a SQLite store for review items and traces, so pending reviews survive a restart.

Step 4: Stage-level evaluation and fixes

- Compared the graph with the legacy pipeline claim by claim. On simple claims they matched (coverage 0.28 vs 0.29, accuracy 0.90 vs 0.88), so the port was faithful. Multi-part claims were the problem.
- **Fix 1, sub-claim aggregation.** Previously one noisy "false" sub-claim made the whole claim FALSE. Now supported plus refuted sub-claims give MIXED_EVIDENCE, and refutations only win when nothing is supported.
- **Fix 2, NLI judge.** Isolated test of the NLI stage on 2,259 claim/evidence pairs (section 6). Replaced mDeBERTa with a Qwen judge, keeping a local NLI model as a fallback if the LLM server is unreachable.
- Held-for-review claims are reported as their own category.

5. Full-pipeline results

Same 2,799 claims, same order, every run. "Strict" counts an abstention as a wrong answer; "answered" looks only at claims that received a Real or Fake verdict.

Run	Coverage	Strict acc.	Strict F1	Answered acc.
Old baseline (no LLM, risk fallback; not comparable)	n/a	0.6831	0.6709	n/a
Legacy pipeline + LLM	61.5%	0.4623	0.4769	0.7515
Graph v1: worst sub-claim wins, mDeBERTa NLI	41.7%	0.3023	0.3124	0.7256
Graph v2: new aggregation rule	45.9%	0.3669	0.3839	0.7992
Graph v3: + Qwen NLI judge (current)	**71.5%**	**0.5970**	**0.6085**	**0.8347**

Coverage (share of claims given a verdict)



Accuracy on answered claims



Bars drawn to scale (100% = 350 px). Graph v3 is highest on both measures.

What the answered claims look like

Run	Fake claims answered	Fake recall	Real recall	Real called Fake
Legacy + LLM	518 / 907	57.3%	82.8%	207
Graph v3	**684 / 907**	**96.9%**	**76.5%**	**310**

The new pipeline catches far more misinformation but also flags more true statements as false. On the claims each pipeline answered, always guessing "Real" would score 0.699 (legacy) and 0.658 (graph v3), so both are above that floor: 0.752 and 0.835.

Verdict distribution (Graph v3)

FALSE	SUPPORTED	MIXED_EVIDENCE	INSUFFICIENT_EVIDENCE	SUPPORTED_WITH_CAVEATS	NOT_A_FACT_CHECK	Held for review
973	863	366	168	166	117	146 (5.2%)

INSUFFICIENT_EVIDENCE fell from 1,119 (graph v2) to 168 once the NLI judge replaced mDeBERTa. Excluding the 146 held claims, strict accuracy is 0.630 and coverage 75.5%.

6. NLI stage test

To isolate the NLI stage, each model was given a claim and its gold evidence quote (2,259 rows from the public datasets in the candidate pool: ai4bharat, SciFact, HealthVer, COVID-Fact, PUBHEALTH) and asked for SUPPORTS, REFUTES or NEI. Majority-class guessing scores about 44%.

Judge	Accuracy	Behaviour
mDeBERTa-v3-base (previous)	40.4%	Answers NEI for 57% of rows when only 22% truly are
XLM-RoBERTa-large-xnli	43.0%	Same pattern
DeBERTa-v3-large (mnli/fever/anli)	45.0%	Same pattern
Qwen2.5-7B, single-letter answer with log-probs	58.8%	Overconfident: right 63% of the time when it says 90%+
Qwen2.5-7B, JSON label (adopted)	**65.4%**	**Supports 74%, refutes 69%, NEI 46% recall**



Bars drawn to scale (100% = 350 px).

The dedicated NLI models are trained on short sentence pairs and struggle with abstract-length evidence. The Qwen judge's confidences are interim: each label's confidence is the precision it achieved on these rows (SUPPORTS 0.58, REFUTES 0.79, NEI 0.55) until a calibrator is fitted on the golden set.

7. Infrastructure and how to reproduce

- **LLM server:** Qwen2.5-7B-Instruct-AWQ on vLLM 0.7.2, PBS queue `Gpu1-10g` (job limit one per queue, 48-hour walltime). Runbook: `docs/LOCAL_LLM_DEPLOYMENT.md`.
- **Evaluation jobs:** sharded across 4 processes on the `Gpu4-40g` queue with `scripts/cluster/eval_baseline.pbs`. A 6-shard attempt ran out of GPU memory; 4 fits. Throughput is roughly 18 claims per minute for the legacy path, against about 3 per minute on a laptop.
- The cluster account password is never stored in the repository.

```
# Tests (78 pass)
python -m pytest tests -q

# Validate the golden set
python evaluation/golden_set/validate_golden.py

# Cluster jobs (from the master node, repo synced to ~/FoodFactCheck/TruthCheck_eval)
qsub -v MODE=main  scripts/cluster/eval_baseline.pbs                        # legacy path
qsub -v "MODE=graph,OUT_DIR=evaluation/baselines/my_run" scripts/cluster/eval_baseline.pbs
qsub -v "MODE=nli,NLI_MODELS=default+org/model" scripts/cluster/eval_baseline.pbs

# API with the LangGraph pipeline
uvicorn app.api.main:app --reload       # TRUTHCHECK_NLI_BACKEND=llm|local|auto
```

Results are stored under `evaluation/baselines/`: `phase0_vllm` (legacy + LLM), `step3_graph` (graph v1), `step3_graph_v2_aggregation`, `step4_nli` and `step4_llm_judge` (NLI tests), `step5_graph_llm_nli` (graph v3).

8. Caveats and corrections

- **Correction to an earlier claim.** During this work I said the LLM's own knowledge produced the baseline verdicts. That was wrong: verdicts come from the calibrator over retrieved evidence. The LLM affected only claim decomposition and legacy risk fields.

- **Style artifact (found 25 October).** The 2,799-claim set can be classified at 98.7% from text style alone; see the box in section 1.
- **Evaluation set limits.** The 2,799 claims are sentences scraped from articles, not clean standalone facts, and they overlap the knowledge base. Real/Fake labels were assigned at sentence level. Treat scores as relative.
- **The legacy "fast path"** (strong match to the 50 verified facts) always produced SUPPORTED evidence regardless of the NLI result and scored only 0.66, close to always guessing "Real". It was deliberately not ported to the graph.
- **Possible leakage.** Some of the public datasets used for the NLI test may be in the training data of the models tested. The NLI comparison is a screening test.
- **Scoring choices.** Held-for-review claims count as abstentions in the headline scores. Sub-claim aggregation and the NLI judge were chosen on these runs; both must be re-checked on the reviewed golden set.
- **Interim parts.** Calibration weights are hand-set; the harm score is keyword-based; the knowledge base and the verified-facts file are still separate retrieval sources.

9. Status and next steps

Item	Status	Notes
Step 0: housekeeping	Done	Repo installs on Python 3.11; tests pass
Step 1: honest evaluation	Done	Abstentions visible in all scores
Step 2: golden set	In progress	Candidate pool exists; needs relabelling to 8 verdicts, Hinglish and Hindi claims, evidence quotes, double annotation
Step 3: real components in graph	Done	Postgres storage and retiring the legacy path not done
Step 4: stage evaluation	Partly done	NLI stage measured; retrieval recall needs gold evidence linked to the knowledge base
Step 5: calibration	Not started	Fit confidence and the NLI label probabilities on the golden set
Step 6: error analysis	Not started	Review the 310 real claims flagged false
Step 7: scale and multimodal	Not started	OCR, audio, Postgres/pgvector

1. **Golden set (teammate).** Move the imported public-dataset rows to `evaluation/external/`; build `golden_v1.csv` by hand with 100+ Hinglish and Hindi claims first. Run the validator before every push.
2. **Retrieval stage test** once gold evidence is linked to knowledge-base passages.
3. **Re-run the NLI comparison on Hinglish and Hindi**; the LLM judge depends on glossing quality and the cluster server being up.
4. **Calibrate** confidences and verdict thresholds on reviewed data (Step 5).
5. **Inspect the false positives** (true statements flagged false) before any user-facing release.

10. Commit history

Commit	Change
`42fbebb`	Step 0 cleanup: installable repo, ignore rules, removed stray files
`a0c83d4`	Step 1: abstention-aware scoring, honest evaluation report
`7b10ce2`	Golden set template and annotation guidelines
`3a2666e`	Phase-0 baseline with LLM connected; sharded evaluation on the cluster
`9ce521b`	Step 3: graph on real retrieval, NLI and risk; interrupt review; persistent storage
`88ce2e5`	Step 4: stage NLI evaluation, LLM NLI judge, better sub-claim aggregation

Generated from the repository state at the commit above. Numbers are taken from `evaluation/baselines/*/merged_summary.txt` and the NLI summary files.